Quickstart Guide: zztab2fig v0.2.1

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This quickstart guide covers the key features in zztab2fig v0.2.1. For complete documentation, see the "Object Types and Themes" and "Advanced Features" vignettes.

Installation

```r
# Install from GitHub
# devtools::install_github("rgt47/zztab2fig")
```

```r
library(zztab2fig)
#> zztab2fig 0.2.1 - LaTeX table generation for R
```

Basic Usage

Convert a data frame to a cropped PDF table:

```r
t2f(mtcars[1:6, 1:4], filename = "basic_table", sub_dir = output_dir)
```

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

This generates three files:

- `basic_table.tex` - LaTeX source
- `basic_table.pdf` - Cropped PDF (primary output for document inclusion)
- `basic_table_full.pdf` - Full PDF (before cropping)

Interactive Terminal Graphics (zzvim-R)

When working in a kitty-compatible terminal (Kitty, Ghostty, WezTerm, iTerm2) with zzvim-R, you can preview tables directly in the terminal using `show_table()`:

```r
# Display table in terminal with inline preview
show_table(mtcars[1:6, 1:4])

# With theme
show_table(mtcars[1:6, 1:4], theme = "nejm")

# Statistical objects work too
```

```r
model <- lm(mpg ~ cyl + hp, data = mtcars)
show_table(model)
```

Navigate through your table history:

```r
show_table(mtcars[1:6, 1:4])
show_table(iris[1:6, ])
show_prev()      # Go back to mtcars table
show_next()      # Forward to iris table
show_history()   # List all graphics (plots and tables)
show()           # Redisplay current
```

Tables and plots share a unified history. Use `show_history("table")` to filter to tables only, or `show_history("plot")` for plots.

Export options:

```r
show_table(model, theme = "nejm")
show_zoom()              # Open PDF in external viewer
show_save("table.pdf")   # Save current as PDF
show_save("table.png")   # Save current as PNG
```

Embedding Tables in Rmd/Quarto

For reproducible documents, use `t2f()` with `graph_include()`:

```r
# In an Rmd/qmd code chunk:
t2f(model, theme = "nejm") |> graph_include()
```

The `t2f()` function returns the path to the generated PDF, which `graph_include()` passes to `knitr::include_graphics()`.

This is the recommended workflow for batch rendering:

1. **Interactive exploration**: Use `show_table()` to preview and compare tables
2. **Final document**: Use `t2f() |> graph_include()` for reproducible embedding

```r
# Interactive session (terminal)
show_table(model, theme = "nejm")
show_table(model, theme = "apa")
show_prev()  # compare themes

# Rmd chunk (batch mode)
t2f(model, theme = "nejm", caption = "Table 1") |> graph_include()
```

Journal Themes

Built-in themes provide journal-specific styling:

```r
t2f(mtcars[1:6, 1:4], theme = "nejm",
    filename = "nejm_table", sub_dir = output_dir)
```

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

Available themes: `"nejm"`, `"apa"`, `"nature"`, `"minimal"`

Set a global theme:

```
t2f_theme_set("nejm")
t2f(mtcars[1:6, 1:4])
t2f(iris[1:6, ])
```

Statistical Objects

t2f works directly with statistical objects:

```
model <- lm(mpg ~ cyl + hp + wt, data = mtcars)
t2f(model, filename = "regression", sub_dir = output_dir,
    include = c("estimate", "std.error", "p.value"))
```

Term	Estimate	Std___Error	p__value
(Intercept)	38.752	1.787	<0.001
cyl	-0.942	0.551	0.098
hp	-0.018	0.012	0.140
wt	-3.167	0.741	<0.001

Supported object types include `lm`, `glm`, `anova`, `aov`, `htest`, `matrix`, `table`, plus rms objects (`ols`, `lrm`, `cph`, `orm`, `Glm`, `psm`) and additional types via broom integration (coxph, survreg, nls, polr, prcomp, kmeans, lmerMod, and more).

Inline Tables for R Markdown

Insert tables directly in R Markdown without LaTeX float positioning:

```
model <- lm(mpg ~ cyl + hp, data = mtcars)
t2f_inline(model,
           width = "3in",
           align = "left",
           caption = "Model Results",
           label = "tab:model",
           caption_position = "above")
```

Or use the convenience function for coefficient tables:

```
t2f_coef(model, caption = "Coefficients")
```

Add visual styling with frames and backgrounds:

```r
t2f_inline(model,
           width = "3in",
           frame = TRUE,
           frame_color = "gray",
           background = "gray!5",
           inner_sep = "4pt")
```

Note: Add `\usepackage{caption}` to your R Markdown YAML for captions.

Model Comparison

Compare multiple models side-by-side:

```r
m1 <- lm(mpg ~ cyl, data = mtcars)
m2 <- lm(mpg ~ cyl + hp, data = mtcars)
m3 <- lm(mpg ~ cyl + hp + wt, data = mtcars)

t2f_regression(
  Model1 = m1, Model2 = m2, Model3 = m3,
  stars = TRUE,
  filename = "model_comparison",
  sub_dir = output_dir
)
```

Term	Model1	Model2	Model3
(Intercept)	37.885* (2.074)	36.908* (2.191)	38.752* (1.787)
cyl	-2.876* (0.322)	-2.265* (0.576)	-0.942 (0.551)
hp		-0.019 (0.015)	-0.018 (0.012)
wt			-3.167* (0.741)
N	32	32	32
R-squared	0.726	0.741	0.843
Adj. R-squared	0.717	0.723	0.826

RMS Package Integration

Full support for Frank Harrell's rms (Regression Modeling Strategies) package, commonly used in biostatistics and clinical trials:

```r
library(rms)
dd <- datadist(mtcars)
options(datadist = "dd")
```

Coefficient Tables with Odds/Hazard Ratios

```r
model <- lrm(am ~ mpg + wt, data = mtcars)
t2f(model, exponentiate = TRUE, filename = "lrm_odds_ratios")
```

ANOVA-Style Chunk Tests

Useful for models with spline terms:

```r
model <- ols(mpg ~ rcs(wt, 4) + cyl, data = mtcars)
t2f(model, output = "anova", filename = "ols_anova")
```

RMS Model Comparison

```r
m1 <- ols(mpg ~ wt, data = mtcars)
m2 <- ols(mpg ~ wt + cyl, data = mtcars)
t2f_rms_compare(Simple = m1, Full = m2, filename = "rms_comparison")
```

RMS methods: `t2f.ols()`, `t2f.lrm()`, `t2f.cph()`, `t2f.orm()`, `t2f.Glm()`, `t2f.psm()`, and `t2f_rms_compare()`.

Extensibility: Any Broom Object

For any object supported by the broom package, use `t2f_tidy()`:

```r
library(survival)
cox_model <- coxph(Surv(time, status) ~ age + sex, data = lung)
t2f_tidy(cox_model, filename = "cox_results")

t2f_tidy(cox_model,
         tidy_args = list(exponentiate = TRUE, conf.int = TRUE),
         filename = "cox_hazard_ratios")
```

This provides an escape hatch for the 100+ object types broom supports.

Captions and Labels

Add LaTeX captions and cross-reference labels:

```r
t2f(mtcars[1:6, 1:4],
    caption = "Motor Trend Car Road Tests (1974)",
    label = "tab:mtcars",
    filename = "captioned_table")
```

Column Alignment

Explicit or auto-detected alignment:

```r
t2f(iris[1:6, ], align = NULL)
t2f(mtcars[1:6, 1:4], align = c("l", "r", "r", "r"))
t2f(mtcars[1:6, 1:4], align = "c")
```

Decimal Alignment

Use siunitx for decimal-aligned columns:

```r
df <- data.frame(
  Item = c("A", "B", "C"),
  Value1 = c(1.5, 123.45, 12.345),
  Value2 = c(0.001, 10.1, 1000.01)
)

t2f(df, filename = "decimal_aligned", sub_dir = output_dir,
    align = list("l", t2f_siunitx(table_format = "3.3"),
                 t2f_siunitx(table_format = "4.2")))
```

Item	Value1	Value2
A	1.500	0.001
B	123.450	10.100
C	12.345	1000.010

Multi-Page Tables

For large tables that span multiple pages:

```r
t2f(mtcars,
    longtable = TRUE,
    caption = "Complete mtcars Dataset",
    filename = "long_table")
```

Quick Reference

Key t2f() Parameters

Parameter	Description	Default
`caption`	LaTeX table caption	NULL
`label`	LaTeX label for cross-refs	NULL
`align`	Column alignment (l/c/r)	NULL (auto)
`longtable`	Multi-page table support	FALSE
`crop`	Crop PDF margins	TRUE
`crop_margin`	Margin size (pts)	10
`theme`	Theme name or object	NULL

Key t2f_inline() Parameters

Parameter	Description	Default
`width`	Figure width (LaTeX units)	NULL
`align`	left, center, right	"center"
`caption`	Uses \captionof (no float)	NULL
`label`	LaTeX cross-reference label	NULL
`caption_position`	"above" or "below"	"above"
`frame`	Draw border around table	FALSE
`frame_color`	Border color (xcolor syntax)	"black"
`background`	Background color (xcolor syntax)	NULL
`inner_sep`	Padding inside frame	"2pt"
`format`	"auto", "pdf", "png"	"auto"

Core Functions

Function	Purpose
`t2f()`	Convert object to PDF table
`t2f_inline()`	Inline tables for R Markdown

Function	Purpose
`t2f_coef()`	Quick coefficient tables
`t2f_regression()`	Compare regression models
`t2f_rms_compare()`	Compare rms models
`t2f_tidy()`	Any broom-supported object
`t2f_theme_set()`	Set global theme
`t2f_theme_get()`	Get current theme
`check_latex_deps()`	Check LaTeX dependencies
`ensure_pdfcrop()`	Install pdfcrop via TinyTeX

Terminal Graphics (zzvim-R)

Function	Purpose
`show_table()`	Display table in terminal
`show_plot()`	Display plot in terminal
`show()`	Redisplay last graphic
`show_prev()`	Previous in history
`show_next()`	Next in history
`show_history()`	List all graphics
`show_zoom()`	Open PDF in viewer
`show_save()`	Export to file
`graph_include()`	Embed in Rmd/qmd (batch mode)

S3 Methods (Base)

Object Type	Method
`data.frame`	`t2f.data.frame()`
`matrix`	`t2f.matrix()`
`lm`	`t2f.lm()`
`glm`	`t2f.glm()`
`anova` / `aov`	`t2f.anova()` / `t2f.aov()`
`htest`	`t2f.htest()`

S3 Methods (rms)

Object Type	Method	Notes
`ols`	`t2f.ols()`	Ordinary least squares
`lrm`	`t2f.lrm()`	Logistic regression (OR)
`cph`	`t2f.cph()`	Cox proportional hazards (HR)
`orm`	`t2f.orm()`	Ordinal regression
`Glm`	`t2f.Glm()`	Generalized linear model
`psm`	`t2f.psm()`	Parametric survival

All rms methods support `output = "coef"` (default) or `output = "anova"` for chunk tests, and `exponentiate = TRUE` for odds/hazard ratios.

S3 Methods (via broom)

Object Type	Package
`coxph`, `survreg`, `survfit`	survival
`nls`	stats
`polr`, `multinom`	MASS, nnet
`prcomp`, `kmeans`	stats
`lmerMod`, `glmerMod`	lme4
`lme`	nlme
`Arima`	stats

Use `t2f_tidy()` for any object type supported by broom.

Theme Presets

Function	Style
`t2f_theme_nejm()`	NEJM (medical journals)
`t2f_theme_apa()`	APA (psychology)
`t2f_theme_nature()`	Nature (scientific)
`t2f_theme_minimal()`	Clean, minimal

System Requirements

- R >= 4.0
- LaTeX distribution (TeX Live, MiKTeX, TinyTeX, or MacTeX)
- `pdflatex` and `pdfcrop` executables
- Optional: ImageMagick (for PNG output and terminal graphics)
- Optional: zzvim-R plugin (for `show_table()` and terminal graphics)
- Optional: jsonlite package (for graphics history)

Check your setup:

```r
check_latex_deps()
```

If pdfcrop is missing with TinyTeX:

```r
ensure_pdfcrop(auto_install = TRUE)
```